Notes: Dubois and Lasio (AER, 2018)

Identifying Industry Margins with Price Constraints: Structural Estimation on Pharmaceuticals

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1 Big picture

- **Question.** How much do French pharmaceutical price regulations actually constrain firms, and what do those constraints do to prices, margins, demand, spending, and welfare?
- **Setting.** The paper studies the French anti-ulcer drug market from 2003 to 2013 and compares it with Germany and the United States, where prices are not directly regulated.
- **Core challenge.** The econometrician does not observe the effective price cap imposed by the French regulator. A fall in observed prices could reflect a binding cap, but it could also reflect ordinary demand or cost forces.
- **Main conceptual contribution.** The paper combines a random-coefficient demand system with a supply model of Bertrand pricing under *unobserved price constraints*. The model backs out which products are genuinely constrained and how much margins would be overstated if regulation were ignored.
- **Main mechanism.** French regulation affects both the *demand side* and the *supply side*. The demand side is affected by the TFR maximum-reimbursement rule, which increases out-of-pocket price exposure above the reimbursement ceiling. The supply side is affected by direct price-cut rules and broader regulatory pressure in price approval by the CEPS.
- **Empirical design.** First, the paper estimates demand in France, Germany, and the United States using IMS data and a random-coefficient logit model. Second, it estimates French margins and marginal costs using the unconstrained markets plus cross-market cost restrictions. Third, it simulates counterfactual equilibria under free pricing and under external reference pricing.
- **Baseline reduced-form fact.** Triple-difference regressions show that drugs subject to the French TFR or price-cut rules become significantly cheaper in France after the regulation is introduced.
- **Main structural finding.** Some but not all products are significantly constrained, and the number of constrained products rises sharply in later years, especially after repeated price-cut reforms.
- **Main pricing and margin result.** Ignoring French price regulation would substantially overstate price-cost margins. Once regulation is accounted for, branded margins are materially lower and some spillovers appear on drugs not directly targeted by the rule.
- **Main counterfactual result.** Relative to a free-pricing benchmark, current French regulation generated only modest savings—about 2% of total anti-ulcer expenditures over 2003–2013—but it shifted demand toward branded drugs and away from generics.
- **Main policy comparison.** An external-reference-pricing counterfactual would lower prices further, raise generic penetration, and generate larger savings and larger short-run consumer surplus, but it may compress generic margins so much that long-run survival becomes a concern.
- **Economic takeaway.** Regulation matters not only because it lowers prices directly, but also because it changes competitive interactions across branded and generic drugs. Evaluating pharmaceutical regulation therefore requires a structural equilibrium approach, not just reduced-form price comparisons.

2 Data and measurement (key objects)

2.1 Regulatory background in France

- French pharmaceutical prices are regulated by the **Economic Committee for Health Products (CEPS)**.
- On the **demand side**, France introduced the **TFR** maximum reimbursement price in 2003. If a drug's price is above the TFR ceiling, the patient pays the difference out of pocket.
- On the **supply side**, the regulator introduced explicit **price cuts** after generic entry or after generics had been present for long enough. In the anti-ulcer market, the main price-cut waves occurred in 2006, 2008, 2009, and 2012.
- In 2004, France also introduced **external reference pricing** as a criterion in price setting for some new drugs, based on prices in Italy, Germany, Spain, and the United Kingdom.
- TFR and direct price cuts are mutually exclusive.

Interpretation. The paper studies a market where regulation can operate through different margins: reimbursement exposure for patients, direct pressure on manufacturers' prices, and broader regulatory screening of acceptable price levels.

2.2 Why the anti-ulcer market is a useful laboratory

- The market is the WHO **A02B** anti-ulcer category.
- It contains three main subclasses:
  - **H2 antagonists** (older first-generation drugs),
  - **PPI** proton-pump inhibitors (the dominant newer generation),
  - **Prostaglandins** (a residual, small category).
- The paper argues that there are no close substitutes from other therapeutic classes, so the market can be cleanly defined.
- The market also contains both branded products and a large wave of generic entry after major patent expirations, which makes it ideal for studying how regulation affects brand-generic competition.

Interpretation. The market is narrow enough to model structurally, but rich enough to contain meaningful substitution patterns, generic diffusion, and multiple regulatory interventions.

2.3 Data sources and sample construction

- The main data come from **IMS Health** and cover wholesale transactions in the retail and hospital sectors.
- Sample period: **2003–2013**.

- Countries used for estimation: **France, Germany, and the United States**.
- Additional price data from **Italy, Spain, and the United Kingdom** are used as instruments and for the external-reference-pricing counterfactual.
- Advertising data come from **IMS Health Global Promotional Track** and include detailing expenditures.
- Additional drug characteristics come from the French health agency **HAS**, including indications, side effects, SMR, ASMR, dates of introduction, and TFR status.
- In France over 2003–2013, the sample contains **103 drugs** sold by **29 firms**, of which **13** are branded firms and **16** are generic manufacturers.

Interpretation. The data are unusually rich because the paper observes prices, quantities, advertising, therapeutic characteristics, and comparable products across both regulated and unregulated markets.

2.4 Key measurement objects

Each drug-country-year observation is identified by the drug name, manufacturer, molecule, therapeutic form, and brand type. The main variables are:

- **Quantity:** total sales in standardized units.
- **Price:** average wholesale revenue per standard unit, in constant US dollars.
- **Brand type:** branded versus generic.
- **Drug characteristics:** number of formats, number of side effects, and key therapeutic indications.
- **Advertising:** especially **detailing**, which is direct physician contact by pharmaceutical representatives.
- **Reimbursement regime:** whether the drug is subject to TFR and whether the market price lies above the reimbursement ceiling.

Interpretation. The paper wants demand to depend on both medical quality and institutional price exposure. The TFR variable is particularly important because the same nominal price has a different meaning once the patient must pay the excess above the reimbursement ceiling.

2.5 Main dependent variables

Different parts of the paper use different outcomes:

- In the reduced-form triple-difference exercise, the dependent variable is **log price**.
- In the structural demand model, the data are drug-level **quantities** and implied **market shares**.
- In the supply model, the key latent objects are **marginal costs**, **price-cost margins**, and **Lagrange multipliers** on price constraints.

- In the counterfactuals, the outcomes are **prices**, **quantities**, **total expenditures**, **producer profits**, and **consumer surplus**.

Interpretation. The paper is not just estimating one reduced-form elasticity. It is building a full equilibrium system and then using that system to answer policy questions.

2.6 Descriptive facts

The market evolves strongly over the sample period:

- The number of drugs marketed in France rises from **25** in 2003 to **94** in 2013.
- Generic market share by volume rises from **2%** in 2003 to **82%** in 2013.
- Total quantity sold rises from roughly **1.13 million** to **1.89 million** standard units (in the paper's thousands-of-units scale).
- Average wholesale price per standard unit falls from **0.75** US dollars to **0.32** US dollars.
- Total manufacturer revenue in the class falls substantially over time.

Interpretation. The market becomes much more generic and much cheaper over time, but that evolution is exactly what creates the need to separate pure competitive forces from regulatory pressure.

3 Models and empirical specifications (all equations + interpretation)

3.1 Reduced-form preview: triple-difference on prices

Before turning to the structural model, the paper estimates a reduced-form triple-difference regression of the form

$$\begin{aligned}\log p_{jct} = \alpha + \theta X_{jct} + \delta_1 \mathrm{TFR}_j + \delta_2 \mathrm{PriceCut}_j + \delta_3(\mathrm{TFR}_j \times After_{jt}) + \delta_4(\mathrm{PriceCut}_j \times After_{jt}) \\ + \delta_5(\mathrm{TFR}_j \times After_{jt} \times France_c) + \delta_6(\mathrm{PriceCut}_j \times After_{jt} \times France_c) + FE + \varepsilon_{jct}.\end{aligned} \tag{1}$$

Here the key coefficients are the France-specific post-regulation interactions.

Interpretation. This regression is not the final estimator. It is a reduced-form check showing that regulated drugs become cheaper in France relative to similar drugs in Germany and the United States after the policy is introduced.

3.2 Supply without regulation

If demand is known and prices are unconstrained, firm f solves

$$\Pi_{ft} = \sum_{j \in F_f} (p_{jt} - c_{jt})\, q_{jt}(p_t, a_t) - a_{jt}, \tag{2}$$

where F_f is the set of products owned by firm f, p_{jt} is price, c_{jt} is marginal cost, q_{jt} is quantity, and a_{jt} is detailing expenditure.

The first-order condition for any product j of firm f is

$$q_{jt} + \sum_{k \in F_f} (p_{kt} - c_{kt}) \frac{\partial q_{kt}(p_t, a_t)}{\partial p_{jt}} = 0. \tag{3}$$

In matrix form, this gives the standard markup equation

$$D_f(p_t - c_t) = -\left[D_f Q_{pt} D_f\right]^{-1} D_f q_t. \tag{4}$$

Interpretation. Once demand is estimated, unconstrained price-cost margins can be read directly from the usual Bertrand system.

3.3 Supply with price constraints

In France, firms may face price ceilings. Branded drugs may satisfy

$$p_{jt} \leq \bar{p}_{jt}, \tag{5}$$

while generic drugs are capped as a fraction of the branded price:

$$p_{jt} \leq \tau_{b(j)t} p_{b(j)t}, \tag{6}$$

where $b(j)$ is the branded counterpart and the allowed generic fraction changes over time (55% before 2006, then 50%, 45%, and 40%).

The constrained problem is

$$\max_{\{p_{jt}\}_{j \in F_f}} \sum_{j \in F_f} (p_{jt} - c_{jt}) q_{jt}(p_t) \quad \text{s.t. } p_{jt} \in \Omega_{jt}. \tag{7}$$

The first-order condition becomes

$$q_{jt} + \sum_{k \in F_f} (p_{kt} - c_{kt}) \frac{\partial q_{kt}(p_t)}{\partial p_{jt}} = \lambda_{jt}, \tag{8}$$

where $\lambda_{jt} \geq 0$ is the shadow value of the price constraint.

Interpretation. A positive λ_{jt} means the regulation is binding. The larger is λ_{jt}, the further the observed price is pushed below the unconstrained optimum.

3.4 Identification of margins under regulation

The paper identifies constrained margins using two ideas.

First, it observes unconstrained markets:

$$\lambda_t = 0 \qquad \text{for } t \in S, \tag{9}$$

where S contains the United States and Germany.

Second, it imposes a cross-market cost restriction:

$$c_{jt} - c_{jt'} = (z_{jt} - z_{jt'})'\delta + \omega_{jt}, \qquad E[\omega_{jt} \mid z_{jt} - z_{jt'}] = 0. \tag{10}$$

In the empirical implementation, the paper uses a log-cost restriction:

$$\log c_{jt}^{FR} = \rho_j + \rho_{US} \log c_{jt}^{US} + \rho_{GR} \log c_{jt}^{GR} + \omega_{jt}. \tag{11}$$

The vector of Lagrange multipliers is estimated by constrained nonlinear least squares:

$$\hat{\lambda}_t = \arg\min_{\lambda \geq 0} \left\| \left[I - z_t(z_t' z_t)^{-1} z_t' \right] \log c_t(\lambda, p_t, q_t) \right\|^2 . \tag{12}$$

Interpretation. The idea is simple but powerful: use the same molecules in free-pricing countries to infer what French marginal costs should look like, and then ask how much extra shadow pressure is needed for French observed prices to fit the constrained first-order conditions.

3.5 Demand model

The paper estimates a random-coefficient logit demand system. Utility of consumer i from drug j in year t is

$$u_{ijt} = \sum_k \alpha_{ki} x_{kjt} - \beta_{0i} p_{jt} \mathbf{1}\{p_{jt} \leq \bar{p}_{jt}^{TFR}\} - \beta_{1i} p_{jt} \mathbf{1}\{p_{jt} > \bar{p}_{jt}^{TFR}\} + \zeta_{jt} + \varepsilon_{ijt}, \tag{13}$$

with outside option utility

$$u_{i0t} = \varepsilon_{i0t}. \tag{14}$$

The market share for product j is

$$s_{jt}(x_t, p_t, \zeta_t) = \int s_{ijt}(x_t, p_t, \zeta_t)\, \phi(\nu_i)\, d\nu_i. \tag{15}$$

Random coefficients allow heterogeneity in price sensitivity and in preferences for product characteristics such as brand status.

Interpretation. The crucial institutional feature is that the price coefficient changes when the drug price exceeds the TFR reimbursement ceiling. The same posted price can generate much stronger demand sensitivity once patients must pay the excess themselves.

3.6 Demand instruments and estimation choices

The paper estimates the demand model by simulated method of moments using 500 Halton draws. Price and detailing are instrumented with:

- hospital prices in France and foreign countries,
- foreign prices residualized on molecule, country, and year effects,

- exchange-rate interactions with firm dummies,
- country-specific wages and pharmaceutical price indices,
- and the average price of detailing.

The paper also calibrates market size because only quantities, not total potential market size, are observed. The implied outside-good share falls from about **38%** in 2003 to about **17%** by 2010 and then stabilizes.

Interpretation. The authors deliberately use a flexible but static demand system: rich enough to capture brand-generic substitution and heterogeneous price sensitivity, but not a dynamic learning model.

3.7 Counterfactual equilibrium problems

The paper simulates two main counterfactuals.

Free pricing for branded drugs. Branded products choose prices freely, but generics remain capped relative to the branded version:

$$p_{jt} \leq \tau_{b(j)t} p_{b(j)t} \quad \text{only if } j \text{ is generic,} \tag{16}$$

with all prices constrained to stay above marginal cost.

External reference pricing. Branded prices are capped by a reference ceiling based on other European countries:

$$p_{jt} \leq \bar{p}_{jt}^{ref} \quad \text{if } j \text{ is branded,} \tag{17}$$

and generic caps continue to depend on the branded price.

Interpretation. These counterfactuals let the paper compare the current French system with two alternatives: much weaker supply-side regulation and a stricter external-price benchmark.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the constrained-margin estimates

The key identifying variation comes from comparing the same drugs across:

- a regulated market (France), and
- unconstrained markets (Germany and the United States).

Demand is first estimated in all countries. Then French marginal costs and constraint multipliers are pinned down by requiring those costs to line up with costs in the unconstrained markets once observable cross-market cost shifters are accounted for.

Interpretation. The paper does not identify French regulation from French data alone. The extra leverage comes from observing how the same molecules behave where prices are free.

4.2 Why the paper goes beyond the triple-difference design

The triple-difference exercise is useful, but it cannot tell whether:

- a lower price in France reflects a truly binding cap,
- or whether the firm would have chosen a similar price anyway because of demand and cost conditions.

The structural model adds exactly that missing step by recovering:

- the unconstrained first-order conditions,
- the shadow value of the price cap,
- and the counterfactual equilibrium prices under alternative rules.

Interpretation. Reduced-form evidence says "prices fell." The structural approach says *which products were constrained, by how much, and with what equilibrium spillovers.*

4.3 Demand-side endogeneity and the IV strategy

Prices and detailing are potentially correlated with unobserved demand shocks. The paper addresses this using:

- foreign and hospital prices,
- cost shifters,
- exchange-rate variation,
- and optimal-IV approximations in the spirit of Reynaert and Verboven.

Interpretation. The demand estimates are only credible if the paper can separate willingness to pay from the fact that firms may set higher prices or spend more on detailing when a drug is intrinsically attractive.

4.4 What the paper can and cannot distinguish

The paper is transparent about several limits.

- The estimated "preferences" reflect the joint outcome of patient demand, physician prescribing, pharmacist substitution, and reimbursement incentives. They are not pure patient tastes.
- The model is **static**. It does not model dynamic learning on the demand side or dynamic entry, R&D, and generic survival on the supply side.
- Welfare is therefore interpreted as short-run surplus under the revealed-preference system that the insurance and pharmacy institutions generate.

Interpretation. The paper is strong on short-run equilibrium pricing under regulation; it is weaker on long-run innovation or dynamic market structure effects.

4.5 Relation to the literature

The paper contributes to three literatures at once:

- pharmaceutical demand estimation with differentiated products,
- price regulation and generic competition,
- and empirical IO methods for markets with unobserved inequality constraints.

Interpretation. Its distinctive contribution is methodological: it shows how standard BLP-style demand estimation can be combined with cross-market cost restrictions to recover margins when prices may be capped but the cap itself is not observed.

5 Tables and figures

5.1 Table 1: French anti-ulcer regulation over time

Read:

- The table separates demand-side and supply-side regulation.
- The key demand-side event is the TFR maximum reimbursement rule in 2003, extended in 2006.
- The key supply-side events are successive price-cut reforms in 2006, 2008, 2009, and 2012.
- The targeted molecules are mainly the big branded PPI products once generic entry becomes relevant.

Economic message. The paper's identification strategy rests on the fact that French regulation changed in several observable steps, but the effective tightness of the price ceiling still has to be inferred structurally.

TABLE 1—REGULATORY CHANGES AFFECTING THE ANTI-ULCER MARKET IN FRANCE SINCE 2003

Date	Description
Panel A. Regulatory changes affecting the demand side	
September 2003	Introduction of maximum reimbursement price (TFR) for Cimetidine (Tagamet) and Ranitidine (Zantac and Raniplex)
January 2006	Introduction of TFR on Famotidine (Pepcidine)
Panel B. Regulatory changes affecting the supply side	
January 2006	Price cut of all drugs in a class if generics entered or have been available for 24 months—Omeprazole (Losec) and Lansoprazole (Takepron, Lanzor)
January 2008	Price cut for branded drugs if enough sales of generics of the same or close substitute molecules—Esomeprazole (Nexium)
January 2009	Revision of the price cut rule (18 months instead of 24) for Omeprazole, Lansoprazole, and Pantoprazole (Pantozol)
January 2012	Revision of the price cut rule at generic entry for Omeprazole, Lansoprazole, Pantoprazole, and Rabeprazole (Pariet)

Note: Price cuts in 2006, 2009, 2012 are of different percentages.

5.2 Tables 2 and 3: market evolution and reduced-form price evidence

Read:

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TABLE 2—SUMMARY STATISTICS FOR FRANCE

	Number of drugs			Quantity	Market share (%)		Wholesale Price/std unit		Firm revenue
Year	All	Branded	Generics	(1,000 std units)	Branded	Generics	(US$)	(€)	(1,000 US$)
2003	25	12	13	1,131,140	98	2	0.75	0.55	1,414,127
2004	39	12	27	1,233,735	86	14	0.78	0.57	1,455,683
2005	40	12	28	1,334,331	77	23	0.70	0.51	1,423,278
2006	42	12	30	1,444,926	73	27	0.63	0.46	1,404,320
2007	51	12	39	1,508,936	70	30	0.60	0.44	1,364,369
2008	49	12	37	1,535,426	61	39	0.56	0.41	1,240,799
2009	59	12	47	1,596,513	55	45	0.52	0.38	1,183,251
2010	62	12	50	1,669,252	50	50	0.46	0.33	1,069,219
2011	74	12	62	1,732,595	39	61	0.43	0.31	974,594
2012	86	12	74	1,820,351	29	71	0.39	0.28	853,985
2013	94	11	83	1,890,575	18	82	0.32	0.23	619,035

Notes: Price is the average price per standard unit in US$ or in € (constant exchange rate for 2014:II: €/US$ = 1.3772). Expenses is the total manufacturer-level revenue of the class. Wholesale prices are at the manufacturer level. Here, the market shares are relative to the total sales of anti-ulcer drugs (no outside good as in the model).

regulator introduced price cuts on branded drugs since 2006. Furthermore, reimbursement rules changed over time and affected some drugs with a maximum reim-

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TABLE 3—TRIPLE DIFFERENCE REGRESSION OF LOG PRICES

Variables	log price (1)	log price (2)	log price (3)	log price (4)
Drug characteristics				
Branded	1.648	1.667	1.824	2.023
	(0.484)	(0.507)	(0.501)	(0.597)
Nb. side effects	−0.352	−0.349	−0.341	−0.285
	(0.0957)	(0.103)	(0.0855)	(0.0350)
Formats	0.131	0.123	0.0345	0.0172
	(0.166)	(0.174)	(0.175)	(0.199)
Helicobacter indication	3.199	3.192	2.821	1.790
	(0.960)	(1.062)	(0.197)	(0.679)
NSAID indication	−0.642	−0.477	1.227	−3.531
	(0.440)	(0.483)	(0.356)	(0.645)
Drug group dummies				
Group TFR	0.158	0.198	−0.183	−0.0559
	(0.364)	(0.421)	(0.444)	(0.425)
Group Price Cut	0.166	0.0600	−0.299	−0.750
	(0.280)	(0.290)	(0.295)	(0.321)
Drug group dummies after				
TFR × after	0.341	0.324	0.639	0.468
	(0.558)	(0.622)	(0.690)	(0.724)
Price Cut × after	0.232	0.290	0.113	0.659
	(0.233)	(0.188)	(0.186)	(0.354)
Reimbursement rule change in France				
TFR × after × France	−0.770	−0.814	−1.656	−1.627
	(0.541)	(0.542)	(0.694)	(0.772)
Price regulation in France				
Price Cut × after × France	−0.773	−0.737	−0.293	−0.328
	(0.353)	(0.338)	(0.291)	(0.384)

- Table 2 shows an enormous rise in generic penetration: from 2% of market volume in 2003 to 82% in 2013.
- Quantity rises strongly over the same period, while average price and manufacturer revenue fall.
- Table 3 shows that the France-specific regulation interactions are negative: the price-cut interaction is about −0.77 log points in the simpler specifications, and the TFR interaction is also negative and significant.
- The reduced-form evidence therefore points to strong price effects of French regulation.

Economic message. These two tables motivate the structural exercise. The market clearly changes a lot over time, and regulation seems to matter, but the triple-difference design alone cannot tell how much of the price movement reflects binding caps versus equilibrium responses.

5.3 Tables 4 and 5: demand estimates and substitution patterns

Read:

- In France, the price coefficient is much more negative when price is above the TFR ceiling (−18.73) than when it is below the ceiling (−6.95), meaning demand becomes far more price sensitive once patients pay out of pocket.
- Detailing enters positively, and branded drugs are valued more than generics.
- Average own-price elasticity in France is about −3.6, with branded drugs more elastic than generics on average (−5.7 versus −2.9).
- Cross-price elasticities are strongest within subclasses and especially among PPI molecules, which confirms that substitution is meaningful but far from perfect.

Economic message. The demand system is rich enough to capture both medical differentiation and institutional price exposure. That is crucial because regulation affects not just supply-side margins but also the demand slope facing firms.

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TABLE 4—ESTIMATION RESULTS OF RANDOM COEFFICIENT LOGIT MODEL

	France		Germany		US	
	Mean	Sigma	Mean	Sigma	Mean	Sigma
Price			−5.50	3.00	−4.06	2.97
			(1.50)	(1.87)	(1.36)	(1.52)
Price below TFR ($p_{jt} \times \mathbf{1}_{\{p_{jt} \leq \overline{p}_{jt}^{tfr}\}}$)	−6.95	6.62				
	(2.70)	(3.25)				
Price above TFR ($p_{jt} \times \mathbf{1}_{\{p_{jt} > \overline{p}_{jt}^{tfr}\}}$)	−18.73	6.83				
	(6.24)	(1.38)				
Detailing	0.27		0.17		0.05	
	(0.11)		(0.04)		(0.14)	
Branded	2.44	1.07	2.81	0.27	6.31	3.10
	(1.37)	(3.17)	(1.39)	(0.80)	(4.91)	(1.55)
Nb. formats	0.49		0.38		1.40	
	(0.30)		(0.25)		(0.55)	
Generic × nb. formats	1.90		1.48		0.02	
	(0.48)		(0.29)		(0.99)	
Nb. side effects	−0.25		0.02		−0.39	
	(0.07)		(0.19)		(0.14)	
Generic × nb. side effects	−0.05		−0.22		0.57	
	(0.19)		(0.19)		(0.79)	
Helicobacter indication	1.36		1.73		3.58	
	(0.26)		(0.42)		(0.78)	
NSAID indication	0.61		0.46		0.18	
	(0.29)		(0.16)		(0.35)	
GERD indication	−1.44		−1.52		−1.58	
	(0.35)		(0.25)		(0.42)	
Year fixed effects	Yes		Yes		Yes	
Molecule fixed effects	Yes		Yes		Yes	

Notes: Standard errors in parentheses under each coefficient. The Mean columns report the mean of random coefficients and the estimates of fixed coefficients. The Sigma columns report the estimates of the standard deviation of random coefficients.

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TABLE 5—OWN- AND CROSS-PRICE ELASTICITIES FOR MAIN DRUGS, 2009 (*France*)

	Branded						Generic	
Subclass	H2	H2	PPI	PPI	PPI	Prost.	H2	PPI
Company	Aptalis	Glaxo	AstraZ	AstraZ	Takeda	Pfizer	Mylan	Mylan
Molecule	Cimet.	Ranit.	Omep.	Esom.	Lanso.	Miso.	Ranit.	Omep.
Drug name	Tagamet	Zantac	Losec	Nexium	Takepron	Cytotec		
Tagamet	−6.96	2.84	0.003	0.23	0.05	0.01	0.02	0.14
Zantac	0.71	−5.62	0.002	0.19	0.04	0.01	0.02	0.13
Losec	0.00	0.00	−5.48	1.64	0.16	0.00	0.003	0.05
Nexium	0.00	0.00	0.19	−3.90	0.25	0.005	0.01	0.14
Takepron	0.00	0.00	0.13	1.73	−5.24	0.01	0.01	0.16
Cytotec	0.002	0.01	0.01	0.81	0.17	−2.45	0.02	0.16
Ranit. Mylan	0.001	0.003	0.02	0.68	0.13	0.01	−3.00	0.23
Omep. Mylan	0.001	0.002	0.04	0.91	0.16	0.01	0.03	−3.68

Notes: Each column is the price elasticity of demand for the drug in the first row with respect to the drug named in the first column. Company names: Glaxo is GlaxoSmithKline. AstraZ is AstraZeneca. Molecules: Ranit. is Ranitidine; Omep. is Omeprazole; Esom. is Esomeprazole; Lanso. is Lansoprazole; Miso. is Misoprostol.

cost of the drug for the patient is increased by the difference between p_{jt} and $\overline{p}_{jt}^{tfr}$; thus, it is understandable that patients are much more price sensitive. The heterogeneity in price sensitivity is however similar across the different reimbursement regimes as suggested by the similar magnitude of the sigmas associated with the two price coefficients.

From this estimated demand model, the mean own-price elasticity across products and years for France is −3.6 and ranges from −13 to −1 across products. Overall, generics show lower own-price elasticities than branded drugs (−2.9 versus

5.4 Tables 6 and 7: binding constraints and price-cost margins

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TABLE 6—NLLS ESTIMATES OF λ_{jt} (*Normalized by Market Size*)

	All products		Products with λ_{jt} significantly positive				
	Number of products		Number of products		λ_{jt}	s_{jt}	$\sum_j s_{jt}$
Year	Branded	Generics	Branded	Generics	Mean	Mean	Mean
2003	12	11	1	5	0.00026	0.00105	0.00628
2004	12	27	1	9	0.00029	0.00183	0.01826
2005	12	28	1	3	0.00027	0.00084	0.00334
2006	12	30	2	5	0.00023	0.01168	0.08176
2007	12	37	3	10	0.00033	0.00904	0.11756
2008	12	36	1	6	0.00030	0.00127	0.00891
2009	12	47	0	11	0.00031	0.00138	0.01519
2010	12	50	1	11	0.00083	0.00318	0.03816
2011	12	62	2	26	0.00152	0.00634	0.17764
2012	12	74	4	36	0.00210	0.00626	0.25034
2013	11	83	3	47	0.00288	0.00858	0.42907

for each product.[11] Table 6 reports the constrained nonlinear least squares estimates

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TABLE 7—AVERAGE PRICE-COST MARGINS BY MOLECULE (*France*)

		Not accounting for price regulation (%)			Accounting for price regulation (%)		
Subclass	Molecule	All drugs	Branded	Generic	All drugs	Branded	Generic
H2	Cimetidine	52	15	62	41	14	47
	Ranitidine	30	24	32	23	19	24
	Famotidine	28	10	36	20	9	26
	Nizatidine	18	18		14	14	
PPI	Omeprazole	29	22	30	22	14	23
	Esomeprazole	43	24	50	31	15	37
	Lansoprazole	39	20	45	29	14	34
	Pantoprazole	43	20	47	33	14	36
	Rabeprazole	45	18	60	33	13	43
Prost.	Misoprostol	47	47		37	37	
Combi.	Bismuth/ Antibiotic	27	27		21	21	

Notes: Margins as a percentage of price. Empty cells indicate that there is no generic version of the molecule named in the corresponding row. When not taking into account price regulation, we infer price cost margins as if firms were free to choose prices.

Read:

- Table 6 shows that not all drugs are constrained, but the number of significantly constrained products rises sharply over time.
- In 2003, the constrained products account for only about 0.6% of market share; by 2013 they account for about 42.9%.
- Table 7 shows that ignoring regulation materially overstates margins. For example, all-drug margins for Omeprazole fall from 29% to 22%, Esomeprazole from 43% to 31%, and Rabeprazole from 45% to 33% once regulation is taken seriously.
- Branded margins end up in the rough range of 9%–37%, while generic margins are often higher, around 23%–47%.

Economic message. The point is not just that prices are lower under regulation. It is that equilibrium margins and inferred marginal costs look different once one recognizes that firms may be sitting on the boundary of a regulator-imposed price cap.

5.5 Tables 8 and 9: counterfactual savings and welfare

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TABLE 8—COUNTERFACTUAL SAVINGS AND SURPLUS FROM FREE PRICING

		2003	2006	2009	2012	2003–2013
Subclass	Molecule					
H2	Cimetidine	−913	−239	−28	−83	−2,120
	Ranitidine	−894	−578	−384	1133	−4,007
	Famotidine	−37	8	18	42	134
	Nizatidine	−4	1	3	6	5
PPI	Omeprazole	−469	1,916	2,111	13,091	59,149
	Lansoprazole	666	616	−468	3,853	9,973
	Pantoprazole	82	333	1,710	1,203	17,495
	Esomeprazole	1,672	−1,067	1,095	4,488	44,213
	Rabeprazole	407	389	−832	28,273	27,918
Prost.	Misoprostol	−277	−239	−181	−145	−2,375
Combi.	Bismuth/ Antibiotic					53
Total		234	1,140	3,044	51,860	150,439
Subtotal	Branded	382	1,903	4,107	30,389	88,242
	Generics	−148	−763	−1,064	21,471	62,197
Consumer surplus change		+1.1%	+1.3%	+2.6%	+10.2%	+15.4%

Notes: Savings are in 1,000 US$. Negative numbers indicate increased expenditures compared to observed. Surplus change is as a percentage of the surplus under the current regulation.

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TABLE 9—COUNTERFACTUAL SAVINGS AND SURPLUS WITH EXTERNAL REFERENCE CAPS

		2004	2006	2009	2012	2004–2013
Subclass	Molecule					
H2	Cimetidine	−608	−365	−103	675	−652
	Ranitidine	−339	−1,323	−951	4,534	2,525
	Famotidine	83	−114	−78	95	−61
	Nizatidine	−4	−15	21	31	38
Subtotal H2						
PPI	Omeprazole	−6,241	−17,175	−6,751	43,880	34,593
	Lansoprazole	−292	4,997	−2,596	25,622	81,427
	Pantoprazole	4,985	8,565	4,663	20,743	109,294
	Esomeprazole	1,210	851	5,180	9,162	68,689
	Rabeprazole	1,531	4,340	3,995	21,829	99,573
Subtotal PPI						
Prost.	Misoprostol	45	−363	−153	637	−113
Combi.	Bismuth/ Antibiotic					382
Total		371	−601	3,226	127,209	395,695
Subtotal	Branded	7,594	20,025	24,687	27,216	267,745
	Generics	−7,222	−20,626	−21,462	99,993	127,950
Consumer surplus change		+6%	+9%	+24%	+30%	+39%

Notes: Savings are in 1,000 US$. Negative numbers indicate increased expenditures relative to observed. Surplus

Read:

- Table 8 shows that current French regulation generates about **$150.4 million** in savings over 2003–2013 relative to the free-pricing benchmark, which is about **2%** of total expenditures in the class.
- Under current regulation, consumer surplus is about **15.4%** higher than under the free-pricing counterfactual.
- Table 9 shows that external reference pricing would generate much larger total savings—about **$395.7 million** over 2004–2013.
- External reference pricing also raises short-run consumer surplus by about **39%**, and it shifts demand more clearly toward generics.

Economic message. The current French system lowers prices, but only modestly. A stricter external-price benchmark would save more money and raise short-run surplus more, yet it would do so by squeezing generic margins very hard, which raises longer-run concerns.

6 Short summary

- The paper studies French anti-ulcer drug regulation using a structural demand-and-supply model.
- Demand is estimated with a random-coefficient logit using France, Germany, and the United States.
- Supply is modeled as Bertrand pricing with possible, unobserved price caps in France.
- Identification comes from unconstrained foreign markets plus cross-market cost restrictions.
- Reduced-form evidence shows that regulated drugs get cheaper in France after policy changes.
- Structural estimates show that some products are bindingly constrained, especially in later years and especially among generics.
- Ignoring regulation leads to overstated price-cost margins.

- Current French regulation yields modest savings relative to free pricing and shifts demand toward branded drugs.
- External reference pricing would save more and increase generic penetration, but possibly at the cost of threatening generic viability in the long run.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that French pharmaceutical regulation materially affects equilibrium pricing, but not in a uniform way across all products. Some drugs are genuinely constrained and others are not. Once these hidden constraints are modeled explicitly, estimated margins fall and the implied effects of regulation look more nuanced than reduced-form price regressions suggest.

7.2 Economic intuition

A price cap changes not only the price of the drug it directly targets. It also changes the equilibrium environment for nearby substitutes. In a differentiated pharmaceutical market, that means regulation has *spillovers*: branded prices, generic prices, and market shares all move together. The resulting expenditure and welfare effects depend on how strong brand preferences are, how close branded and generic substitutes are, and how the reimbursement system changes effective price sensitivity.

7.3 Takeaway

The main lesson is that pharmaceutical price regulation cannot be evaluated with simple before-after price comparisons. The right object is the **counterfactual equilibrium**: which prices firms would have chosen without the rule, how substitutes respond, and how those price changes reallocate demand and profit across branded and generic producers.